

Assignment 5: Linear and Nonlinear Systems

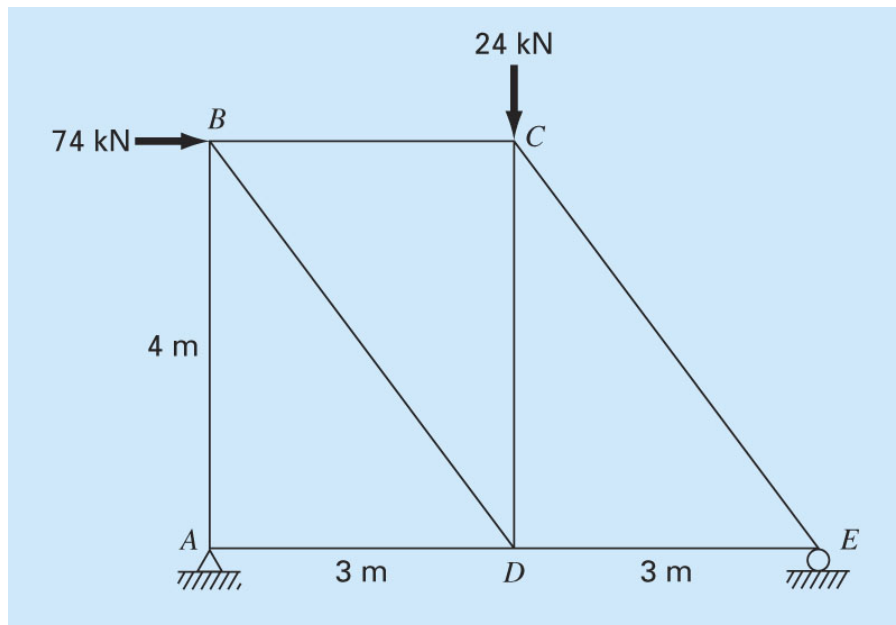
1. A civil engineer involved in construction requires 4800, 5800, and 5700 m³ of sand, fine gravel, and coarse gravel, respectively for a building project. There are three pits from which these materials can be obtained. The composition of these pits is:

	Sand %	Fine Gravel %	Coarse Gravel %
Pit 1	55	30	15
Pit 2	25	45	30
Pit 3	25	20	55

Set up a linear system to solve for the volume of material (in cubic meters) that must be hauled from each pit in order to meet the engineer's needs. You can use the '\ ' operator to solve for the linear system.

2. A truss is loaded as shown in the figure below. Using force balance equations at each node in the horizontal and vertical directions and/or moment equations about *A* and *E*, solve for the 10 unknowns (7 member forces and reactions at *A* and *E*). Note that there is no horizontal reaction at *E*. Use the following notation and assume tensile member forces are positive.

$F_1=F_{AB}$, $F_2=F_{AD}$, $F_3=F_{BD}$, $F_4=F_{BC}$, $F_5=F_{CD}$, $F_6=F_{DE}$, $F_7=F_{CE}$, $F_8=R_{AX}$, $F_9=R_{AY}$, $F_{10}=R_{EY}$. Print the results by sorting the absolute values. i.e., if *x* is your solution vector then print `sort(abs(x))`.



3. The pH of water has great significance to environmental and chemical engineers. It can be related to processes ranging from pipe corrosion to acid rain. The pH is related to hydrogen ion concentration by

$$pH = -\log_{10}[H^+]$$

The following five equations govern the concentrations of a mixture of carbon dioxide and water for a closed system:

$$K_1 = \frac{[H^+][HCO_3^-]}{[CO_2]}$$

$$K_2 = \frac{[H^+][CO_3^{2-}]}{[HCO_3^-]}$$

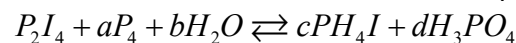
$$K_w = [H^+][OH^-]$$

$$c_T = [CO_2] + [HCO_3^-] + [CO_3^{2-}]$$

$$Alk = [HCO_3^-] + 2[CO_3^{2-}] + [OH^-] - [H^+]$$

Where Alk = the alkalinity, c_T = total inorganic carbon, and the K 's are equilibrium coefficients. The five unknowns are $[CO_2]$ = carbon dioxide, $[HCO_3^-]$ = bicarbonate, $[CO_3^{2-}]$ = carbonate, $[H^+]$ = hydrogen ion, and $[OH^-]$ = hydroxyl ion. Solve for the five unknowns given that $Alk = 2 \times 10^{-3}$, $c_T = 3 \times 10^{-3}$, $K_1 = 10^{-6.3}$, $K_2 = 10^{-10.3}$, and $K_w = 10^{-14}$. Also, calculate the solution's pH . Use your own Newton Raphson's implementation and Matlab's `fsolve` function and compare solutions. You may want to choose the levenberg-marquardt algorithm for `fsolve` with an initial guess of 10^{-3} for all unknowns. Set the tolerance for `fsolve` at 10^{-12} .

4. Implement a function for Gauss-Jordan method with partial pivoting to solve a system of linear equations. When balancing the following chemical reaction by conserving the number of atoms of each element between the reactants and products:



The unknown stoichiometric coefficients a , b , c , and d are given by the solution of the following system of equations (by balancing the number of atoms for each P , I , H , and O):

$$\begin{bmatrix} -4 & 0 & 1 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & -2 & 4 & 3 \\ 0 & -1 & 0 & 4 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ 0 \end{bmatrix}$$

- (i) Solve for the stoichiometric coefficients using your Gauss-Jordan function compare your results with the `\'` operator.
- (ii) Modify your Gauss-Jordan function to calculate the inverse of an $n \times n$ square matrix. Use this code to calculate the inverse of the above 4×4 matrix and compare your results with the MATLAB's `inv` function.